ACTA BIOMEDICA 2026; VOL. 97, N. 4: 18782 DOI: 10.23750/ABM.2026.18782 Mattioli 1885

ORIGINAL ARTICLE

Occupational workload and cardiovascular risk among commercial airline pilots

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ABSTRACT

Background and aims: Commercial airline pilots are exposed to cumulative occupational stressors, including circadian rhythm disruption and prolonged sitting, which may increase cardiovascular disease (CVD) risk. This study aimed to analyze the association between body mass index (BMI) and total flight hours with cardiovascular disease risk among commercial airline pilots.

Methods: A cross-sectional study was conducted involving 226 pilots. Cardiovascular risk was assessed using the Framingham Risk Score (FRS). Data were analyzed using Spearman correlation and ordinal logistic regression to identify independent predictors.

Results: The majority of respondents were classified as obese (65.9%), and 50.0% were categorized as having moderate cardiovascular risk. Bivariate analysis showed a significant association between BMI and cardiovascular risk ($p = 0.035$; $r = 0.140$) and between total flight hours and cardiovascular risk ($p = 0.001$; $r = 0.269$). In multivariable analysis, total flight hours (AOR = 1.56; 95% CI: 1.18–2.07; $p = 0.002$) and age (AOR = 1.09; 95% CI: 1.03–1.15; $p = 0.004$) remained significant predictors, while BMI was no longer significant ($p = 0.094$).

Conclusion: Total flight hours are an independent predictor of cardiovascular disease risk among commercial airline pilots, whereas BMI does not independently predict risk after adjustment. (www.actabiomedica.it)

Key words: Body Mass Index, cardiovascular risk, flight hours, Framingham risk score, pilots

Received: 24/02/2026 | Accepted: 25/04/2026

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Introduction

Cardiovascular diseases constitute a group of disorders involving the heart and blood vessels, including coronary heart disease, cerebrovascular disease, hypertension, and other vascular conditions, and they remain the leading cause of mortality worldwide (1–5). These diseases are chronic and progressive in nature and often develop gradually without clear symptoms in the early stages, resulting in late detection. Cardiovascular disease risk is influenced by various factors, including non-modifiable factors such as age, sex, and genetic predisposition, as well as modifiable factors such as dietary patterns, physical activity, smoking habits, blood pressure, and nutritional status as reflected by body mass index. The complexity of these risk factors positions cardiovascular disease as a major public health problem that requires comprehensive preventive strategies and early detection (6).

Body mass index is a widely used indicator of nutritional status for assessing the risk of metabolic and cardiovascular diseases (7). Overweight and obesity are known to contribute to dyslipidemia, hypertension, insulin resistance, and chronic inflammatory processes that play a crucial role in the pathogenesis of cardiovascular disease. In addition, occupational factors also influence cardiovascular risk, particularly in professions characterized by high physical and psychological demands, irregular work schedules, and chronic stress exposure (8,9). In the aviation context, pilots represent a professional group with unique occupational characteristics that may increase cardiovascular disease risk, thereby requiring special attention from the perspectives of occupational health and flight safety (10).

Globally, the World Health Organization reports that cardiovascular diseases account for nearly one-third of all deaths worldwide each year, with the majority resulting from coronary heart disease and stroke (11). In Indonesia, data from the Basic Health Research Survey indicate a continuous increase in the prevalence of cardiovascular diseases, alongside lifestyle changes and a rising prevalence of risk factors such as obesity and hypertension (12). Among aviation personnel, particularly pilots, several international studies have reported a high prevalence of overweight, obesity, and other cardiovascular risk factors associated with long working hours, sleep disturbances, and insufficient physical activity (13). At the national level, routine medical examinations of pilots conducted at Aviation Medical Centers also reveal a considerable proportion of pilots with cardiovascular disease risk, especially among those with high cumulative flight hours (1,2,14).

This phenomenon highlights a discrepancy between the professional demands placed on pilots, which require optimal physical fitness and health, and the reality of high exposure to cardiovascular risk factors within the aviation work environment (15,16). On one hand, pilots are expected to maintain excellent health to ensure flight safety; on the other hand, occupational characteristics such as intensive flight schedules, long-haul flights, time zone changes, and limited opportunities for physical exercise may increase health risks. This condition creates a phenomenon gap between the expected health standards and the actual health status of some commercial airline pilots, particularly with regard to cardiovascular disease risk (17).

Although several occupational mechanisms such as circadian rhythm disruption, cosmic ionizing radiation exposure, prolonged sedentary behavior, and occupational stress have been proposed to explain increased cardiovascular risk among pilots, these mechanisms are not directly measurable within routine aeromedical examination data. Therefore, cumulative total flight hours are used in this study as a proxy indicator of long-term occupational exposure. While this approach allows practical assessment using available data, it may not fully capture the complexity and biological pathways underlying cardiovascular risk in this population.

In addition to this phenomenon, a research gap also warrants attention. Most studies on cardiovascular disease focus on the general population or specific occupational groups, while research specifically examining cardiovascular risk factors among commercial airline pilots in Indonesia remains limited (18). Previous studies have often emphasized single factors, such as flight hours or physical activity, without simultaneously considering nutritional status indicators such as body mass index (19). A multidimensional approach is therefore required to achieve a more comprehensive understanding of cardiovascular disease risk among professional groups with unique occupational characteristics, such as pilots (20).

Research on the relationship between body mass index, total flight hours, and cardiovascular disease risk is particularly important given its implications not only for pilots' individual health but also for overall flight safety. Cardiovascular disease may lead to sudden incapacitation during flight, posing serious risks to passengers, cabin crew, and flight operations. Consequently, identifying factors that contribute to increased cardiovascular disease risk among pilots is of high urgency as a foundation for developing preventive strategies, health surveillance programs, and aviation safety policies.

The novelty of this study lies in its effort to simultaneously examine the association between body mass index and total flight hours with cardiovascular disease risk among commercial airline pilots in Indonesia using a standardized risk assessment approach. This study also provides a contextual overview based on data from pilots' medical examinations conducted at aviation health facilities, making the findings highly relevant for decision-making in the fields of aviation medicine and occupational health. Thus, this research not only contributes to the scientific body of knowledge but also holds strong practical applicability.

Based on the above considerations, this study aims to determine the association between body mass index and total flight hours with cardiovascular disease risk among commercial airline pilots in Indonesia. The findings are expected to provide theoretical benefits by contributing to the development of knowledge in occupational health and cardiovascular science, as well as practical benefits by informing aviation health institutions in designing preventive programs and health monitoring strategies for pilots. In addition, this study is expected to serve as a reference for future research related to pilot health and flight safety.

Methods

This study employed an analytical observational design with a cross-sectional approach using retrospective secondary data extracted from routine aeromedical medical records. The retrospective nature of the study implies that all variables were obtained from pre-existing clinical documentation rather than collected prospectively, which may introduce information bias and limit control over data completeness and measurement consistency. Despite these limitations, this design allows efficient analysis of standardized medical examination data within a defined occupational cohort. The study was conducted at the Aviation Medical Center (Public Service Agency/BLU) in Jakarta, a national referral center for aviation personnel medical examinations. Secondary data were obtained from routine medical examinations conducted between January and June 2024, with data extraction performed in March 2025 following administrative approval.

The study population comprised all commercial airline pilots who underwent mandatory periodic medical examinations during the study period. The target population included actively employed scheduled commercial airline pilots aged 40–65 years holding a valid Air Transport Pilot License (ATPL) and having accumulated at least 1,500 total flight hours. Pilots were included if complete medical record data were available, including anthropometric measurements, blood pressure, lipid profile, smoking status, and diabetes history. Pilots with a documented history of cardiovascular disease (such as coronary artery disease or stroke), those receiving antihypertensive or lipid-lowering therapy, or those with incomplete key variable data were excluded to avoid distortion of cardiovascular risk assessment. A purposive sampling approach was applied by including all eligible pilots who met the predefined criteria. Based on Slovin's formula with a 5% margin of error, the minimum required sample size was 226 participants, and all eligible cases were included to enhance statistical precision.

The primary outcome variable was cardiovascular disease risk assessed using the Framingham Risk Score (FRS), calculated based on age, sex, systolic blood pressure, total cholesterol, HDL cholesterol, smoking status, and diabetes status. The FRS was categorized into low, moderate, and high cardiovascular risk levels according to established guidelines. The main exposure variables were BMI and total cumulative flight hours. BMI was calculated as weight in kilograms divided by height in meters squared using measurements obtained by trained healthcare professionals with calibrated equipment and categorized according to World Health Organization criteria. Total flight hours were obtained from official flight log records documented in medical files and categorized into low (1,500–4,999 hours), moderate (5,000–9,999 hours), and high (≥10,000 hours).

To address potential confounding, additional variables were included in the analysis, including age, sex, smoking status, blood pressure, lipid levels, diabetes status, and where available, indicators of physical activity, sleep quality, occupational stress, alcohol consumption, and dietary patterns documented during health screening. Rather than relying solely on inclusion and exclusion criteria, confounders were controlled analytically through multivariate modeling.

Data were extracted using structured data collection forms and entered into Microsoft Excel, followed by verification and cleaning procedures before statistical analysis using SPSS version 25. Descriptive analysis was conducted to summarize participant characteristics. Normality of continuous variables was assessed using the Kolmogorov–Smirnov test, which indicated non-normal distribution; therefore, non-parametric methods were applied where appropriate. Bivariate associations between ordinal exposure variables and cardiovascular risk categories were assessed using Spearman's rank correlation test. To control for confounding factors and evaluate independent associations, ordinal logistic regression analysis was performed with cardiovascular risk category as the dependent variable, and adjusted odds ratios with 95% confidence intervals were reported. An interaction term between BMI and total flight hours was included in the regression model to assess potential effect modification. Age-stratified analyses were conducted, whereas sex-stratified analysis was not feasible due to the very small proportion of female pilots; therefore, sex was retained as a covariate in the multivariate model. Statistical significance was determined at $p < 0.05$ (two-tailed).

To minimize bias, standardized objective medical examination data were used, uniform data extraction procedures were applied, and eligibility criteria were strictly enforced. The use of a homogeneous occupational cohort further reduced variability related to external population differences, thereby strengthening internal validity.

In addition, this study is subject to the "healthy worker effect," as only actively employed pilots undergoing mandatory medical certification were included. Pilots with severe cardiovascular disease or high-risk profiles may have been excluded from active duty, resulting in a systematic underrepresentation of high-risk individuals. This bias is likely to attenuate the observed associations between occupational exposure and cardiovascular risk.

Results

Based on Table 1, the majority of respondents were male, accounting for 222 individuals (98.2%), while female respondents comprised only 4 individuals (1.8%). The age distribution indicates that the largest proportion of pilots was in the 45–49 year age group, totaling 97 individuals (42.9%), followed by those aged 50–54 years with 64 individuals (28.3%). The smallest age group was 60–63 years, consisting of 6 individuals (2.7%), indicating that most active pilots were within the productive age range.

Table 2 shows that more than half of the respondents were classified as Obesity Class I, with 121 individuals (53.5%), while 28 individuals (12.4%) were classified as Obesity Class II. Pilots with normal BMI accounted for only 26 individuals (11.5%), and the underweight category represented the smallest proportion with 2 individuals (0.9%). These findings indicate

Table 1. Characteristics of commercial airline pilot respondents.

Variable	n	%
Sex		
Male	222	98.2
Female	4	1.8
Age (years)		
40–44	35	15.5
45–49	97	42.9
50–54	64	28.3
55–59	24	10.6
60–63	6	2.7

Table 2. Distribution of Body Mass Index, total flight hours, and cardiovascular disease risk (FRS).

Variable	n	%
Body Mass Index (BMI)		
Underweight	2	0.9
Normal	26	11.5
Overweight	49	21.7
Obesity Class I	121	53.5
Obesity Class II	28	12.4
Total Flight Hours		
Low (1,500–4,999 hours)	8	3.5
Moderate (5,000–9,999 hours)	53	23.5
High (≥10,000 hours)	165	73
Cardiovascular Disease Risk (FRS)		
Low Risk	66	29.2
Moderate Risk	113	50
High Risk	47	20.8

Table 3. Bivariate association between BMI, flight hours, and cardiovascular risk (Spearman Test).

Variable	p-value	r (Spearman)
BMI vs Cardiovascular Risk	0.035	0.14
Total Flight Hours vs Cardiovascular Risk	0.001	0.269

Table 4. Cross-tabulation of BMI and cardiovascular risk.

BMI Category	Low Risk n (%)	Moderate Risk n (%)	High Risk n (%)
Underweight	1 (1.5)	1 (0.9)	0 (0.0)
Normal	13 (19.7)	10 (8.8)	3 (6.4)
Overweight	13 (19.7)	26 (23.0)	10 (21.3)
Obesity Class I	35 (53.0)	58 (51.3)	28 (59.6)
Obesity Class II	4 (6.1)	18 (15.9)	6 (12.8)

Table 5. Cross-Tabulation of total flight hours and cardio vascular risk.

Flight Hours	Low Risk n (%)	Moderate Risk n (%)	High Risk n (%)
Low	6 (75.0)	2 (25.0)	0 (0.0)
Moderate	20 (37.7)	20 (37.7)	13 (24.5)
High	35 (21.2)	91 (55.2)	39 (23.6)

Table 6. Ordinal logistic regression analysis of factors associated with cardiovascular risk.

Variable	Adjusted OR (95% CI)	p-value
BMI (per category increase)	1.18 (0.97–1.44)	0.094
Total Flight Hours (per category increase)	1.56 (1.18–2.07)	0.002*
Age (per 1-year increase)	1.09 (1.03–1.15)	0.004*
Smoking Status (Yes vs No)	1.42 (0.88–2.30)	0.148
Diabetes (Yes vs No)	1.67 (0.91–3.06)	0.097
HDL Cholesterol (per 1 mg/dL increase)	0.97 (0.95–0.99)	0.018*

Table 7. Interaction analysis between BMI and total flight hours.

Variable	Adjusted OR (95% CI)	p-value
BMI × Flight Hours	1.07 (0.89–1.28)	0.472

Table 8. Multicollinearity diagnostics using variance inflation factor (VIF).

Variable	VIF	Interpretation
Age	4.2	Moderate
Flight Hours	4.8	Moderate
BMI	2.1	Low
Smoking	1.6	Acceptable
Diabetes	2.3	Low
HDL	2.8	Low

a high prevalence of overweight and obesity among commercial airline pilots. Regarding total flight hours, the majority of respondents had high total flight hours (≥10,000 hours), totaling 165 individuals (73.0%), whereas pilots with low flight hours accounted for only 8 individuals (3.5%). This finding suggests that most respondents were highly experienced pilots. Based on the Framingham Risk Score, most pilots were categorized as having moderate cardiovascular disease risk, with 113 individuals (50.0%), followed by low risk with 66 individuals (29.2%) and high risk with 47 individuals (20.8%).

Bivariate analysis using Spearman's correlation test revealed a statistically significant association between BMI and cardiovascular disease risk ($p = 0.035$; $r = 0.140$). However, the strength of correlation was very weak. In contrast, total flight hours showed a stronger and statistically significant association with cardiovascular risk ($p = 0.001$; $r = 0.269$), although the magnitude of correlation remained in the weak-to-moderate range (Table 3).

Table 4 presents the cross-tabulation between BMI categories and cardiovascular risk among commercial airline pilots. Overall, Obesity Class I was the most prevalent BMI category across all cardiovascular risk groups, accounting for 53.0% of pilots in the low-risk group, 51.3% in the moderate-risk group, and 59.6% in the high-risk group. The proportion of pilots classified as overweight was relatively similar across risk categories, comprising 19.7% of the low-risk group, 23.0% of the moderate-risk group, and 21.3% of the high-risk group. Pilots with Obesity Class II were more frequently observed in the moderate-risk (15.9%) and high-risk (12.8%) groups than in the low-risk group (6.1%). In contrast, normal BMI was less common among pilots with higher cardiovascular risk, decreasing from 19.7% in the low-risk group to 8.8% in the moderate-risk group and 6.4% in the high-risk group. Underweight pilots represented the smallest proportion in the study, with only 1.5% in the low-risk group, 0.9% in the moderate-risk group, and no underweight participants in the high-risk group.

The cross-tabulation between total flight hours and cardiovascular risk demonstrated a clearer pattern. Among pilots with low flight hours, the majority were categorized as low risk (75.0%). Conversely, among those with high flight hours, most were classified as moderate risk (55.2%) and high risk (23.6%). These findings suggest an increasing trend in cardiovascular risk with greater cumulative flight-hour exposure (Table 5).

Ordinal logistic regression analysis showed that after adjusting for covariates, total flight hours remained an independent predictor of cardiovascular risk (Adjusted OR = 1.56; 95% CI: 1.18–2.07; $p = 0.002$). Age was also significantly associated with increased cardiovascular risk (Adjusted OR = 1.09 per year increase; 95% CI: 1.03–1.15; $p = 0.004$). Meanwhile, HDL cholesterol demonstrated a protective effect against cardiovascular risk (Adjusted OR = 0.97; 95% CI: 0.95–0.99; $p = 0.018$).

BMI was no longer statistically significant after adjustment ($p = 0.094$), suggesting potential confounding effects, particularly by age and metabolic factors. Smoking status and diabetes mellitus showed elevated odds ratios but did not reach statistical significance (Table 6).

The interaction analysis between BMI and total flight hours did not show a statistically significant effect ($p = 0.472$). This indicates that the association between BMI and cardiovascular risk did not significantly differ across levels of flight-hour exposure. Therefore, BMI and total flight hours appear to independently influence cardiovascular risk rather than interact synergistically (Table 7).

Multicollinearity diagnostics showed that all independent variables had VIF values below 5, indicating no evidence of severe multicollinearity. The highest VIF values were observed for age (VIF = 4.2) and total flight hours (VIF = 4.8), suggesting moderate correlation but within acceptable limits (Table 8).

Discussion

This study successfully achieved its primary objective, which was to examine the association between body mass index (BMI) and total flight hours with cardiovascular disease (CVD) risk among commercial airline pilots. Using the Framingham Risk Score (FRS) as a standardized cardiovascular risk assessment tool, this study was able to identify the distribution of cardiovascular risk levels and determine whether occupational exposure and anthropometric factors were associated with increased risk. The findings provide a clearer understanding of how both traditional and occupational factors contribute to cardiovascular risk in a safety-critical professional group.

The results demonstrated a high prevalence of overweight and obesity among pilots, with the majority classified as Obesity Class I. In addition, half of the respondents were categorized as having moderate cardiovascular risk. Bivariate analysis showed that both BMI and total flight hours were significantly associated with cardiovascular risk; however, the strength of the association for BMI was weak. After multivariable adjustment, total flight hours, age, and HDL cholesterol remained significantly associated with cardiovascular risk, whereas BMI was no longer statistically significant. No significant interaction effect was observed between BMI and total flight hours, suggesting that these variables contribute to cardiovascular risk through separate pathways rather than synergistically.

The high prevalence of obesity in this study is consistent with previous research on aviation professionals, which has reported elevated rates of overweight and metabolic risk due to sedentary work patterns, irregular schedules, and circadian rhythm disruption (20,21). Several occupational health studies have indicated that long-haul pilots and those with extended flying careers tend to exhibit increased cardiometabolic risk profiles. The finding that cumulative flight hours were independently associated with cardiovascular risk aligns with earlier studies suggesting that long-term occupational exposure may contribute to chronic physiological stress and cardiovascular burden (22).

From a theoretical perspective, the relationship between total flight hours and cardiovascular risk can be interpreted within the cumulative exposure framework (21). Prolonged exposure to circadian rhythm disruption, cosmic ionizing radiation at cruising altitude, limited physical activity during long-haul flights, and chronic occupational stress may contribute to endothelial dysfunction, oxidative stress, and metabolic dysregulation. These mechanisms are biologically plausible pathways linking occupational exposure to increased cardiovascular risk over time. However, it should be emphasized that these mechanisms were not directly measured in this study, and total flight hours were used as a proxy indicator of cumulative exposure (23).

Regarding BMI, although a statistically significant association with cardiovascular risk was observed in the bivariate analysis, the correlation was very weak. The loss of statistical significance in the multivariable model suggests that the relationship between BMI and cardiovascular risk may be influenced by confounding factors, particularly age and metabolic components included in the FRS calculation. In addition, the use of FRS, which incorporates lipid and metabolic parameters, may introduce a degree of circularity that attenuates the independent contribution of BMI. Therefore, BMI in this study may be better interpreted as a general marker of metabolic status rather than a strong independent predictor of cardiovascular risk.

In contrast, total flight hours demonstrated a stronger and consistent association with cardiovascular risk across both bivariate and multivariable analyses. However, given the observational and cross-sectional nature of the study, this finding should be interpreted as an association rather than evidence of causality. The persistence of the association after adjustment suggests that cumulative occupational exposure may play an important role, although residual confounding cannot be ruled out.

Given the conceptual relationship between age and total flight hours, the possibility of multicollinearity was carefully considered. Multicollinearity diagnostics using the Variance Inflation Factor (VIF) indicated that all variables remained within acceptable limits (VIF < 5), suggesting that the regression model was not substantially affected by multicollinearity. Nevertheless, as more experienced pilots tend to be older, some degree of shared variance between these variables is expected, and their independent effects should be interpreted with caution.

Several methodological considerations should also be acknowledged. First, this study used a cross-sectional design, which limits the ability to infer causal relationships. Second, the analysis was based on retrospective secondary data extracted from medical records, which may introduce information bias and limits control over data completeness and measurement consistency. Third, although several covariates were considered, other potentially important confounders such as physical activity, sleep quality, dietary patterns, and occupational stress were not consistently available, raising the possibility of residual confounding.

In addition, the use of the Framingham Risk Score in an aeromedical examination context may introduce selection bias. Pilots with known cardiovascular disease or high risk profiles may be less likely to remain in active duty due to regulatory restrictions, leading to an underrepresentation of high-risk individuals. This phenomenon, commonly referred to as the healthy worker effect, may attenuate the observed associations and should be considered when interpreting the findings.

The implications of this study are relevant for aviation medicine and occupational health practice. Cardiovascular risk assessment in pilots may benefit from incorporating indicators of cumulative occupational exposure, such as total flight hours, alongside traditional risk factors. Preventive strategies should extend beyond weight management and include interventions targeting sleep hygiene, fatigue management, circadian rhythm stabilization, and stress reduction.

Conclusion

This study found that total flight hours, age, and HDL cholesterol were significantly associated with cardiovascular disease (CVD) risk among commercial airline pilots. Although body mass index (BMI) showed a statistically significant association in the bivariate analysis ($p = 0.035$; $r = 0.140$), it was not statistically significant in the multivariable model ($p = 0.094$), suggesting that its effect may be influenced by confounding factors and metabolic components included in the Framingham Risk Score.

The observed association between total flight hours and cardiovascular risk suggests that cumulative occupational exposure may play a role in shaping cardiovascular risk profiles among pilots. However, given the cross-sectional design, the use of retrospective secondary data, and the potential for residual confounding, these findings should be interpreted as associative rather than causal. In addition, the possibility of multicollinearity between age and total flight hours, although within acceptable limits, and the presence of a healthy worker effect may have influenced the magnitude of the observed associations.

These findings highlight the potential importance of considering occupational exposure history, including cumulative flight hours, alongside traditional cardiovascular risk factors in aeromedical assessment. Preventive strategies may benefit from a comprehensive approach that includes cardiovascular screening, fatigue management, promotion of physical activity, and monitoring of metabolic health.

Further longitudinal studies are needed to better clarify causal relationships and to directly assess the impact of occupational exposures on cardiovascular health among airline pilots.

Ethic approval: This study was approved by the Health Research Ethics Committee, Faculty of Medicine, Halu Oleo University, Kendari, Indonesia (Protocol No. 112/UN4.6.4.5.31/PP36/2024).

Conflict of interest: Each author declares that he or she has no commercial associations (e.g., consultancies, stock ownership, equity interest, patent/licensing arrangements, etc.) that might pose a conflict of interest in connection with the submitted article.

Authors contribution: J contributed to conceptualization, study design, data acquisition, data analysis, and manuscript drafting. RBL contributed to study design, data interpretation, and critical revision of the manuscript. MA contributed to cardiology expertise, interpretation of clinical data, and manuscript review. SHA contributed to data management, statistical analysis support, and manuscript editing. All authors read and approved the final manuscript.

Declaration on the use of AI: ChatGPT (OpenAI) was used to assist in language editing and grammatical refinement. All scientific content, data analysis, and interpretations remain the sole responsibility of the authors.

Consent for publication: Not applicable.

Acknowledgments: The authors would like to thank the Aviation Medical Center (BLU) Jakarta for providing access to anonymized medical examination data used in this study.

Funding: This research received no external funding.

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